

Anagh Dave

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Education

University Of Maryland, College Park, MD

Dec 2024

Master of Science, Mechanical Engineering

Manipal Institute of Technology, Manipal, India

May 2022

Bachelor of Technology, Mechanical Engineering

Technical Skills

Simulation Software: ANSYS Fluent, OpenFOAM, MATLAB, SU2, IcePak

CAD Software: Solidworks, SpaceClaim, Autodesk, CATIA, FreeCAD

Programming Language: C++, C, Linux Shell Scripting

Experience

Thermal Test Engineer (Contract) · Microsoft · Mountain View, CA

Jan 2026-Present

- Led system-level thermal, airflow, and acoustic validation of electronic cooling solutions, defining test plans, metrics, and pass/fail criteria aligned with product requirements
- Performed fan characterization (PQ curves), SPL noise measurements, and thermal–acoustic optimization, enabling data-driven cooling design trade-offs
- Designed and executed instrumentation strategies with 20+ thermocouples/RTDs and airflow sensors, achieving highly repeatable datasets (less than 5% variation)
- Conducted thermal root-cause analysis (hotspots, recirculation, fan anomalies) and partnered with design teams to implement corrective actions at the system level
- Evaluated performance of diverse thermal interface materials (TIMs) (greases, polymers, liquid metals, graphene), quantifying the impact of bond line thickness and clamping force on junction temperatures under varying loads

Mechanical Engineering Co-Op · Teradyne · North Reading, MA

Feb 2025 -Dec 2025

- Designed and fabricated a testing fixture for graphene-based thermal interface materials (TIMs), transitioning SolidWorks PDM concepts into integrated hardware with NIDAQ instrumentation to enable precise thermal resistance measurements and reduce testing time by 15 hours per week.
- Developed and prototyped copper microchannel cold plates for AI chip cooling using SolidWorks and FDM 3D printing, improving thermal management efficiency.
- Performed structural analysis on a triangulated fixture support for coolant filters used in ATE systems, verifying stresses remaining below 10% of yield and confirming mechanical robustness.
- Coordinated with external vendors to source custom components and materials for cold plate manufacturing, ensuring design feasibility, cost-effectiveness, and timely delivery.
- Collaborated cross-functionally with test engineers to validate thermal and mechanical performance through hands-on testing and iterative design improvements.
- Designed and validated microchannel cold plates and the supporting coolant loop for high heat flux cooling (300 W/cm²), using CFD and experimental testing to reduce peak device temperature by 10–15°C and guide product-level thermal design decisions.

Graduate Research Assistant · NEIT Lab · University of Maryland · College Park, MD

Sep 2023 – Dec 2024

- Featured as the Invention of the Year at the University of Maryland
- Conducted CFD investigation of innovative Microchannel Heat Sink design with herringbone structures over a cold plate to cool electronics components using ANSYS.
- Implemented Micro-Encapsulated Phase Change Material (MEPCM) slurry flow in microchannels using User Defined Functions (UDFs) in ANSYS Fluent.
- Optimized the microchannel system using Multi-Objective Optimization through formulated regression models using MATLAB.
- Developed a novel optimal design with a 300% increase in heat transfer performance over a regular plain channel system.

Thermal Management of Electronics · Course Project · University of Maryland, College Park, MD

Apr 2023 – May 2023

- Designed and evaluated a novel vortex-enhanced heat sink configuration using punched delta vortex generators to improve thermal performance in pin-fin heat sinks
- Achieved ~5% enhancement in heat transfer through geometric optimization and flow manipulation strategies
- Developed detailed CFD models in ANSYS IcePak, including geometry creation, meshing, and boundary condition setup for accurate thermal-fluid simulation
- Analyzed airflow patterns, pressure drop, and thermal distribution to inform design trade-offs between cooling performance and flow resistance

Summer Intern · Defense Research and Development Laboratory · Hyderabad, India

Jan 2022 – May 2022

- Developed MATLAB-based FEA solver for thermal simulation of defense hardware components under extreme thermal loads
- Mastered manual meshing techniques and applied advanced numerical methods, such as Gauss-Seidel iterations, to model complex heat transfer scenarios.
- Delivered actionable insights by simulating critical thermal conditions, aiding in refining thermal management strategies